Additional file 1

When does fragment size selection improve non-invasive detection of trisomy 21? A simulation study of fetal fraction, sequencing depth and noise

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Contents: Figures S1–S2 (method comparison at no and low noise); Table S1 (B-opt chosen cutoffs); Table S2 (C2 feature importances); Tables S3–S4 (W with true vs. fixed FF); Table S5 (AUC of all methods in all conditions); Table S6 (detection limits with conservative and matched-FPR estimates). All values are computed on the test group of each condition. FF, fetal fraction; FPR, false positive rate; depth is the number of fragments after filtering.

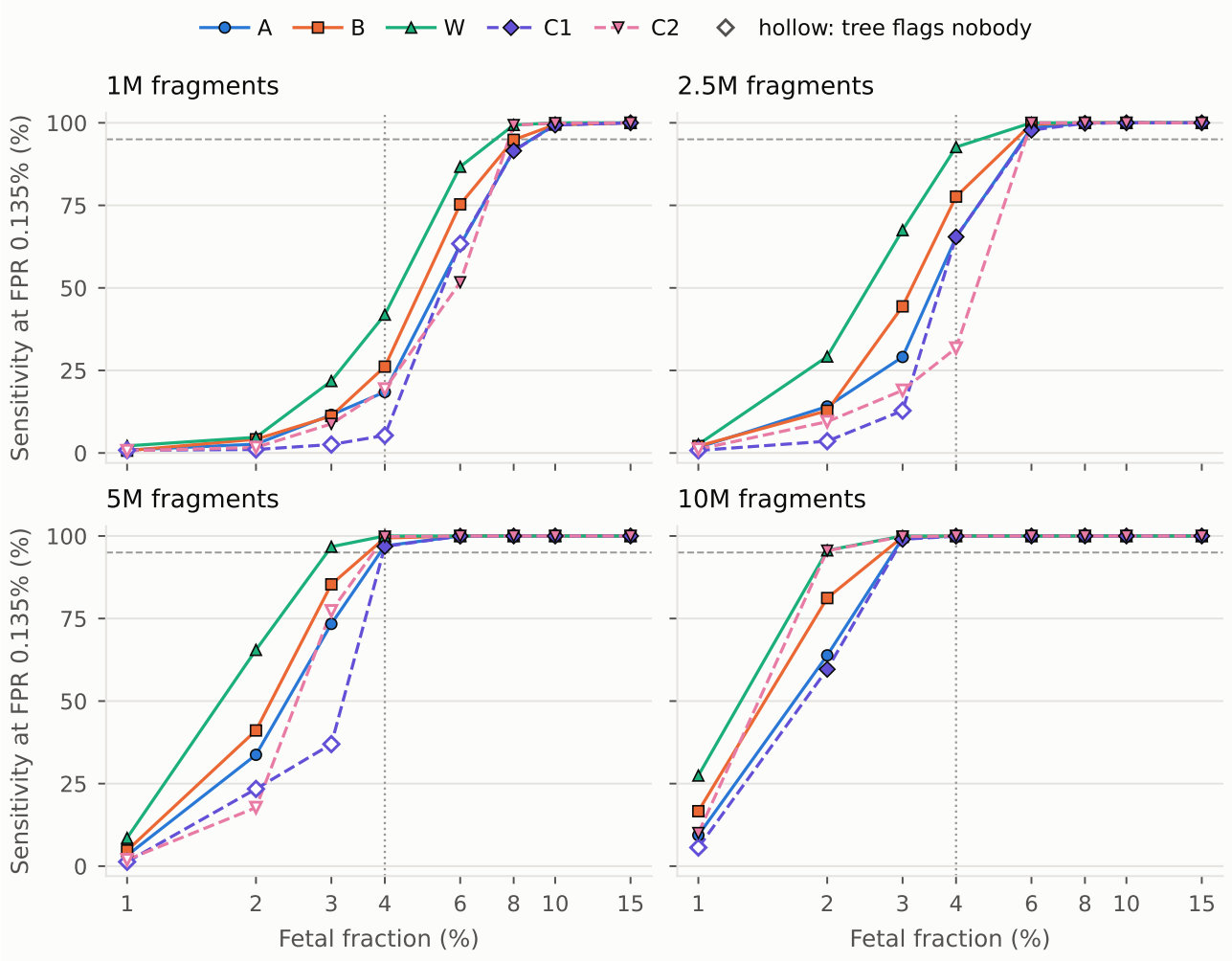


Figure S1. Sensitivity of the five methods with no additional noise ($\sigma = 0$). Shown as in Fig. 5 of the main text: sensitivity at FPR 0.135% against fetal fraction for A, B, W, C1 and C2, one panel per depth. Hollow markers indicate conditions where the tree's calibrated threshold classified no patient as trisomic.

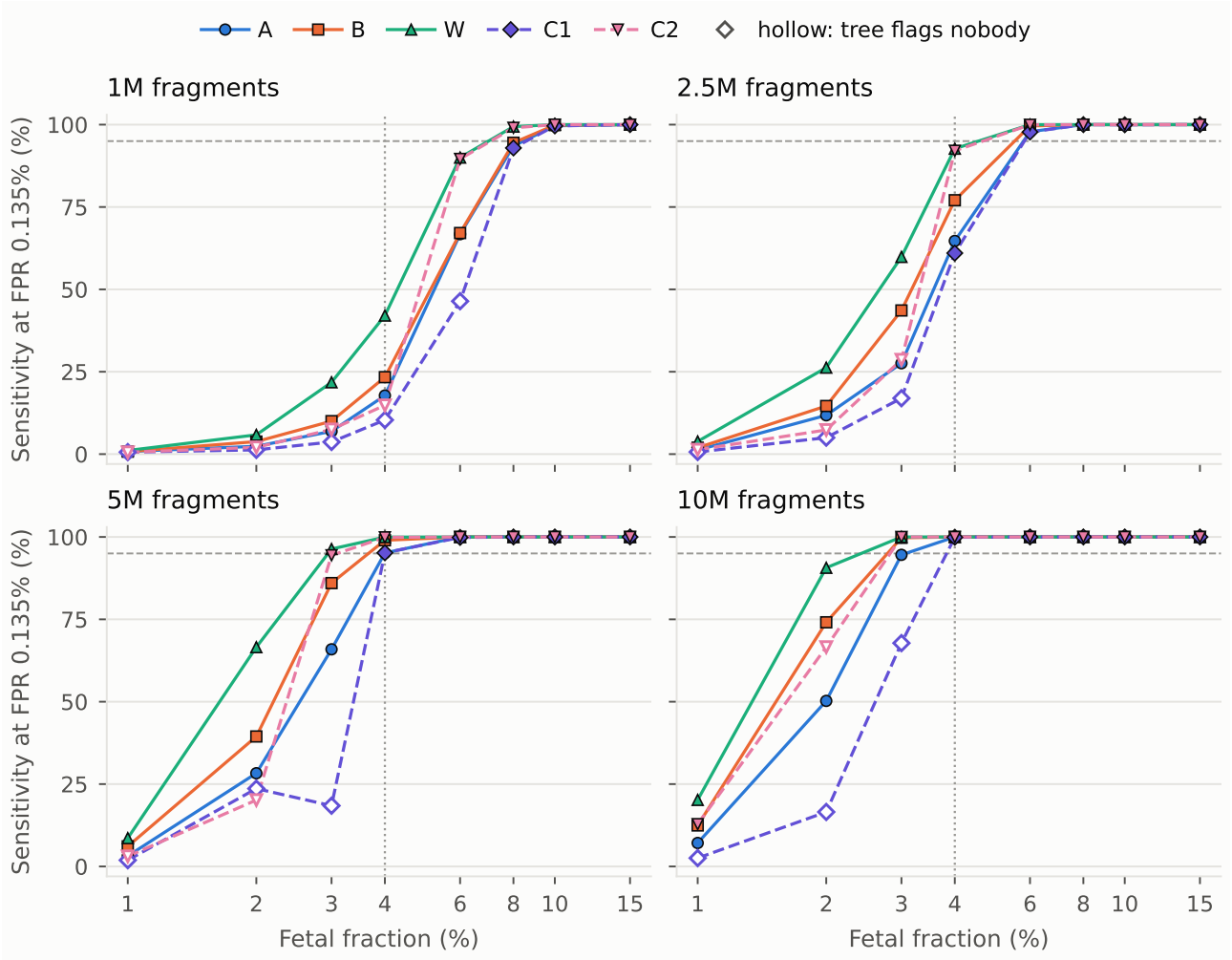


Figure S2. Sensitivity of the five methods at low noise ($\sigma = 0.15\%$). As Figure S1.

Table S1. Cutoff chosen by B-opt in each condition. The upper cutoff was selected on the training group from 120–170 bp in 5 bp steps (lower bound 40 bp). The table gives the training separation, the test separation of B-opt and B (40–156 bp), and the sensitivity of both at FPR 0.135%, with their difference (B-opt minus B, in percentage points). Chosen cutoffs: 145 bp in 4 conditions, 150 bp in 8, 155 bp in 46, 160 bp in 34 and 165 bp in 4. B-opt minus B in sensitivity at FPR 0.135% averaged +0.15 points (range −5.9 to +4.4). Sep., standardised separation; Sens., sensitivity at FPR 0.135%.

Noise	Depth (M)	FF (%)	Cutoff (bp)	Train sep. B-opt	Test sep. B-opt	Test sep. B	Sens. (%) B-opt	Sens. (%) B	Diff. (points)
None	1	1	160	0.61	0.60	0.60	1.0	0.7	+0.3
None	1	2	155	1.21	1.24	1.24	5.0	4.2	+0.8
None	1	3	160	1.81	1.82	1.83	14.4	11.2	+3.2
None	1	4	160	2.46	2.40	2.43	26.6	26.2	+0.4
None	1	6	160	3.61	3.59	3.61	69.8	75.3	−5.5
None	1	8	160	4.67	4.77	4.78	95.0	94.8	+0.2
None	1	10	160	5.79	5.82	5.83	99.6	99.5	+0.1
None	1	15	165	8.44	8.38	8.41	100.0	100.0	+0.0
None	2.5	1	160	0.95	0.98	0.98	1.8	2.1	−0.4
None	2.5	2	155	1.98	1.96	1.95	13.2	12.8	+0.4
None	2.5	3	155	2.85	2.93	2.94	48.0	44.4	+3.6
None	2.5	4	160	3.86	3.82	3.82	78.6	77.6	+0.9
None	2.5	6	155	5.62	5.61	5.62	99.5	99.6	−0.1
None	2.5	8	160	7.56	7.42	7.45	100.0	100.0	+0.0
None	2.5	10	160	9.19	9.37	9.25	100.0	100.0	+0.0
None	2.5	15	165	13.54	13.55	13.37	100.0	100.0	+0.0
None	5	1	155	1.37	1.34	1.35	4.0	4.8	−0.8
None	5	2	155	2.72	2.78	2.78	42.9	41.1	+1.8
None	5	3	155	4.02	4.09	4.10	85.2	85.4	−0.1
None	5	4	160	5.46	5.46	5.48	99.0	99.4	−0.4
None	5	6	160	7.92	7.96	7.94	100.0	100.0	+0.0
None	5	8	155	10.66	10.54	10.54	100.0	100.0	+0.0
None	5	10	160	13.13	13.05	13.02	100.0	100.0	+0.0
None	5	15	165	19.02	19.04	18.90	100.0	100.0	+0.0
None	10	1	155	1.98	1.97	1.98	15.4	16.7	−1.2
None	10	2	155	3.93	3.86	3.87	84.5	81.2	+3.4
None	10	3	155	5.80	5.73	5.75	99.6	99.7	−0.1
None	10	4	160	7.65	7.63	7.67	100.0	100.0	+0.0
None	10	6	160	11.34	11.47	11.40	100.0	100.0	+0.0
None	10	8	155	14.77	15.14	15.15	100.0	100.0	+0.0
None	10	10	160	18.59	18.40	18.26	100.0	100.0	+0.0
None	10	15	165	27.00	26.67	26.41	100.0	100.0	+0.0
Low	1	1	155	0.60	0.62	0.62	0.6	0.8	−0.1
Low	1	2	155	1.22	1.25	1.25	3.8	3.8	+0.0
Low	1	3	160	1.80	1.79	1.81	9.1	10.0	−0.9
Low	1	4	155	2.41	2.43	2.42	27.8	23.4	+4.4
Low	1	6	155	3.59	3.57	3.57	70.0	67.1	+2.9
Low	1	8	160	4.67	4.73	4.75	94.0	94.5	−0.5
Low	1	10	160	5.81	5.83	5.78	99.6	99.9	−0.2
Low	1	15	160	8.33	8.47	8.39	100.0	100.0	+0.0
Low	2.5	1	160	0.93	0.93	0.95	1.4	1.9	−0.5
Low	2.5	2	155	1.95	1.99	1.99	16.6	14.6	+2.0
Low	2.5	3	155	2.87	2.85	2.85	40.6	43.6	−3.0
Low	2.5	4	155	3.86	3.79	3.80	78.3	77.0	+1.3
Low	2.5	6	155	5.57	5.60	5.63	99.5	99.6	−0.1
Low	2.5	8	155	7.45	7.38	7.41	100.0	100.0	+0.0
Low	2.5	10	160	9.26	9.15	9.15	100.0	100.0	+0.0
Low	2.5	15	160	13.33	13.19	13.10	100.0	100.0	+0.0
Low	5	1	155	1.34	1.41	1.41	7.5	6.1	+1.5
Low	5	2	155	2.67	2.69	2.70	39.4	39.5	−0.1
Low	5	3	155	4.00	4.10	4.11	86.2	86.0	+0.2
Low	5	4	155	5.27	5.36	5.35	99.0	99.0	+0.0
Low	5	6	155	7.89	7.95	7.97	100.0	100.0	+0.0
Low	5	8	160	10.43	10.41	10.44	100.0	100.0	+0.0
Low	5	10	160	12.57	12.74	12.80	100.0	100.0	+0.0
Low	5	15	160	18.71	18.50	18.36	100.0	100.0	+0.0
Low	10	1	155	1.89	1.93	1.93	15.0	12.4	+2.5
Low	10	2	155	3.76	3.73	3.74	73.5	74.1	−0.6
Low	10	3	155	5.49	5.66	5.66	99.7	99.8	−0.1
Low	10	4	155	7.57	7.45	7.47	100.0	100.0	+0.0
Low	10	6	155	10.84	10.94	10.97	100.0	100.0	+0.0

Table S1 continued

Noise	Depth (M)	FF (%)	Cutoff (bp)	Train sep. B-opt	Test sep. B-opt	Test sep. B	Sens. (%) B-opt	Sens. (%) B	Diff. (points)
Low	10	8	155	14.28	14.58	14.64	100.0	100.0	+0.0
Low	10	10	160	17.76	17.93	17.86	100.0	100.0	+0.0
Low	10	15	160	26.01	26.17	25.95	100.0	100.0	+0.0
Realistic	1	1	145	0.63	0.56	0.57	0.9	1.1	−0.1
Realistic	1	2	160	1.20	1.20	1.20	2.9	2.5	+0.4
Realistic	1	3	150	1.82	1.83	1.83	11.8	9.0	+2.7
Realistic	1	4	155	2.38	2.37	2.38	25.6	23.8	+1.8
Realistic	1	6	155	3.46	3.50	3.52	62.8	66.9	−4.1
Realistic	1	8	155	4.59	4.63	4.64	93.5	94.2	−0.6
Realistic	1	10	160	5.72	5.76	5.76	99.4	99.4	+0.0
Realistic	1	15	160	8.43	8.33	8.25	100.0	100.0	+0.0
Realistic	2.5	1	150	0.89	0.95	0.96	1.7	1.4	+0.3
Realistic	2.5	2	155	1.94	1.93	1.93	11.5	14.6	−3.2
Realistic	2.5	3	155	2.78	2.80	2.79	45.6	43.0	+2.6
Realistic	2.5	4	155	3.64	3.65	3.65	73.9	72.7	+1.2
Realistic	2.5	6	155	5.44	5.39	5.40	99.2	99.0	+0.2
Realistic	2.5	8	160	7.05	7.08	7.19	100.0	100.0	+0.0
Realistic	2.5	10	160	8.89	8.71	8.77	100.0	100.0	+0.0
Realistic	2.5	15	160	12.88	12.87	12.90	100.0	100.0	+0.0
Realistic	5	1	145	1.32	1.25	1.25	4.5	4.2	+0.3
Realistic	5	2	150	2.54	2.58	2.54	36.4	36.0	+0.4
Realistic	5	3	150	3.86	3.84	3.84	78.5	79.2	−0.8
Realistic	5	4	155	5.08	5.09	5.09	97.9	98.1	−0.2
Realistic	5	6	155	7.39	7.43	7.42	100.0	100.0	+0.0
Realistic	5	8	155	9.66	9.79	9.79	100.0	100.0	+0.0
Realistic	5	10	155	11.97	11.96	11.99	100.0	100.0	+0.0
Realistic	5	15	155	17.55	17.34	17.32	100.0	100.0	+0.0
Realistic	10	1	145	1.71	1.80	1.72	12.3	8.1	+4.3
Realistic	10	2	145	3.48	3.49	3.43	60.0	65.8	−5.9
Realistic	10	3	150	5.25	5.17	5.07	98.2	97.8	+0.4
Realistic	10	4	150	6.73	6.80	6.68	100.0	100.0	+0.0
Realistic	10	6	150	9.86	9.95	9.89	100.0	100.0	+0.0
Realistic	10	8	150	13.16	13.06	12.93	100.0	100.0	+0.0
Realistic	10	10	155	15.90	15.64	15.60	100.0	100.0	+0.0
Realistic	10	15	155	22.78	22.85	22.80	100.0	100.0	+0.0

Table S2. C2 feature importances in each condition. Impurity-based feature importances (scikit-learn) of the three C2 features: z_A, the z-score of method A; z_{short}, the short-fragment z-score; and z_W, the z-score of method W. Max depth is the tree depth selected by cross-validation. Flags nobody: the tree's calibrated threshold classified no patient as trisomic in that condition. At FF $\geq$ 10% all methods detected nearly every case, so the importances there say little about each feature.

Noise	Depth (M)	FF (%)	z_A	z_{short}	z_W	Max depth	Flags nobody
None	1	1	0.012	0.000	0.988	3	Yes
None	1	2	0.000	0.000	1.000	3	Yes
None	1	3	0.004	0.001	0.995	4	No
None	1	4	0.003	0.000	0.997	4	Yes
None	1	6	0.000	0.000	1.000	3	No
None	1	8	0.000	0.000	1.000	3	No
None	1	10	0.000	0.000	1.000	3	No
None	1	15	0.000	0.000	1.000	3	No
None	2.5	1	0.000	0.000	1.000	3	Yes
None	2.5	2	0.000	0.000	1.000	3	Yes
None	2.5	3	0.000	0.000	1.000	3	Yes
None	2.5	4	0.000	0.002	0.998	3	Yes
None	2.5	6	0.000	0.001	0.999	3	No
None	2.5	8	0.000	0.000	1.000	3	No
None	2.5	10	0.000	0.000	1.000	3	No
None	2.5	15	0.000	1.000	0.000	3	No
None	5	1	0.000	0.000	1.000	3	Yes
None	5	2	0.000	0.001	0.999	3	Yes
None	5	3	0.001	0.001	0.999	3	Yes
None	5	4	0.000	0.000	1.000	3	No
None	5	6	0.000	0.000	1.000	3	No
None	5	8	0.000	0.000	1.000	3	No
None	5	10	0.000	1.000	0.000	3	No
None	5	15	0.000	1.000	0.000	3	No
None	10	1	0.001	0.005	0.995	4	No
None	10	2	0.001	0.000	0.999	3	No
None	10	3	0.000	0.000	1.000	3	No
None	10	4	0.000	0.000	1.000	3	No
None	10	6	0.000	1.000	0.000	3	No
None	10	8	0.000	1.000	0.000	3	No
None	10	10	0.000	1.000	0.000	3	No
None	10	15	0.000	1.000	0.000	3	No
Low	1	1	0.000	0.016	0.984	3	Yes
Low	1	2	0.000	0.000	1.000	3	Yes
Low	1	3	0.004	0.004	0.992	4	Yes
Low	1	4	0.000	0.000	1.000	3	Yes
Low	1	6	0.001	0.000	0.999	3	No
Low	1	8	0.000	0.001	0.999	3	No
Low	1	10	0.001	0.000	0.999	3	No
Low	1	15	0.000	0.000	1.000	3	No
Low	2.5	1	0.006	0.000	0.994	3	Yes
Low	2.5	2	0.000	0.000	1.000	3	Yes
Low	2.5	3	0.002	0.000	0.998	3	Yes
Low	2.5	4	0.000	0.003	0.997	4	No
Low	2.5	6	0.000	0.001	0.999	3	No
Low	2.5	8	0.000	0.000	1.000	3	No
Low	2.5	10	0.000	0.000	1.000	3	No
Low	2.5	15	0.000	0.000	1.000	3	No
Low	5	1	0.000	0.000	1.000	3	Yes
Low	5	2	0.000	0.000	1.000	3	Yes
Low	5	3	0.000	0.001	0.999	3	No
Low	5	4	0.000	0.000	1.000	3	No
Low	5	6	0.000	0.000	1.000	3	No
Low	5	8	0.000	0.000	1.000	3	No

Table S2 continued

Noise	Depth (M)	FF (%)	z_A	z_{short}	z_W	Max depth	Flags nobody
Low	5	10	0.000	1.000	0.000	3	No
Low	5	15	0.000	1.000	0.000	3	No
Low	10	1	0.007	0.003	0.991	4	No
Low	10	2	0.001	0.001	0.998	3	Yes
Low	10	3	0.000	0.000	1.000	3	No
Low	10	4	0.000	0.000	1.000	3	No
Low	10	6	0.000	0.000	1.000	3	No
Low	10	8	0.000	1.000	0.000	3	No
Low	10	10	0.000	1.000	0.000	3	No
Low	10	15	0.000	1.000	0.000	3	No
Realistic	1	1	0.020	0.031	0.948	4	Yes
Realistic	1	2	0.000	0.000	1.000	3	Yes
Realistic	1	3	0.000	0.000	1.000	3	Yes
Realistic	1	4	0.000	0.000	1.000	3	Yes
Realistic	1	6	0.000	0.001	0.999	3	Yes
Realistic	1	8	0.000	0.001	0.999	3	No
Realistic	1	10	0.000	0.000	1.000	3	No
Realistic	1	15	0.000	0.000	1.000	3	No
Realistic	2.5	1	0.000	0.000	1.000	3	Yes
Realistic	2.5	2	0.002	0.013	0.985	4	No
Realistic	2.5	3	0.005	0.005	0.990	4	Yes
Realistic	2.5	4	0.006	0.002	0.991	4	Yes
Realistic	2.5	6	0.000	0.000	1.000	3	No
Realistic	2.5	8	0.000	0.000	1.000	3	No
Realistic	2.5	10	0.000	0.000	1.000	3	No
Realistic	2.5	15	0.000	1.000	0.000	3	No
Realistic	5	1	0.022	0.030	0.949	4	Yes
Realistic	5	2	0.017	0.000	0.983	3	Yes
Realistic	5	3	0.003	0.007	0.990	3	No
Realistic	5	4	0.001	0.002	0.996	3	No
Realistic	5	6	0.000	0.000	1.000	3	No
Realistic	5	8	0.000	0.000	1.000	3	No
Realistic	5	10	0.000	0.000	1.000	3	No
Realistic	5	15	0.000	1.000	0.000	3	No
Realistic	10	1	0.031	0.115	0.854	5	Yes
Realistic	10	2	0.024	0.019	0.957	4	No
Realistic	10	3	0.000	0.008	0.991	3	No
Realistic	10	4	0.000	0.000	1.000	3	No
Realistic	10	6	0.000	0.000	1.000	3	No
Realistic	10	8	0.000	1.000	0.000	3	No
Realistic	10	10	0.000	1.000	0.000	3	No
Realistic	10	15	0.000	1.000	0.000	3	No

Table S3. W with weights from the true FF versus weights fixed at FF 10% in each condition. Columns 4–6: W with true-FF weights; columns 7–9: W with weights fixed at FF 10%; columns 10–12: differences (fixed minus true). Sens., sensitivity (%) at the z > 3 threshold; Sens. FPR, sensitivity (%) at FPR 0.135%; differences in sensitivity are in percentage points.

Noise	Depth (M)	FF (%)	Sens.	Sens. FPR	AUC	Sens.	Sens. FPR	AUC	Δ Sens.	Δ Sens. FPR	Δ AUC
None	1	1	2.2	2.1	0.6972	2.1	2.1	0.6960	−0.1	−0.1	−0.0012
None	1	2	8.6	4.8	0.8534	8.3	4.7	0.8522	−0.2	−0.1	−0.0012
None	1	3	23.4	21.8	0.9379	23.2	23.4	0.9376	−0.2	+1.6	−0.0003
None	1	4	49.6	41.9	0.9799	49.3	42.2	0.9796	−0.3	+0.4	−0.0002
None	1	6	90.2	86.7	0.9984	90.0	88.0	0.9984	−0.2	+1.3	+0.0000
None	1	8	99.5	99.4	0.9999	99.5	99.4	1.0000	+0.0	+0.0	+0.0000
None	1	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	1	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	2.5	1	4.5	2.6	0.8043	4.0	3.0	0.8020	−0.4	+0.4	−0.0023
None	2.5	2	25.9	29.2	0.9539	25.0	30.0	0.9523	−0.9	+0.8	−0.0016
None	2.5	3	68.4	67.5	0.9926	68.8	67.8	0.9924	+0.4	+0.2	−0.0001
None	2.5	4	93.2	92.7	0.9992	93.3	91.8	0.9992	+0.1	−0.9	+0.0000
None	2.5	6	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	−0.0000
None	2.5	8	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	2.5	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	2.5	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	5	1	8.0	8.6	0.8717	7.9	9.3	0.8698	−0.1	+0.8	−0.0019
None	5	2	67.0	65.5	0.9908	65.8	63.9	0.9905	−1.1	−1.6	−0.0003
None	5	3	96.7	96.8	0.9994	96.7	96.8	0.9994	−0.1	+0.0	−0.0000
None	5	4	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	5	6	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	5	8	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	5	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	5	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	10	1	23.8	27.5	0.9533	22.8	26.7	0.9515	−1.0	−0.8	−0.0017
None	10	2	95.7	95.7	0.9996	95.2	94.7	0.9996	−0.5	−1.0	−0.0000
None	10	3	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	−0.0000
None	10	4	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	10	6	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	10	8	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	10	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
None	10	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	1	1	1.3	1.1	0.6978	1.2	1.2	0.6956	−0.1	+0.1	−0.0022
Low	1	2	5.8	5.9	0.8514	5.4	5.5	0.8502	−0.4	−0.4	−0.0013
Low	1	3	20.9	21.8	0.9331	20.4	20.8	0.9327	−0.5	−0.9	−0.0004
Low	1	4	47.1	42.0	0.9794	46.7	42.9	0.9791	−0.4	+0.8	−0.0003
Low	1	6	88.6	90.0	0.9990	88.5	90.0	0.9990	−0.1	+0.1	−0.0000
Low	1	8	99.5	99.4	1.0000	99.5	99.5	1.0000	+0.0	+0.1	+0.0000
Low	1	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	1	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	2.5	1	3.8	3.9	0.7891	3.7	3.8	0.7858	−0.1	−0.1	−0.0033
Low	2.5	2	27.6	26.2	0.9481	26.5	25.9	0.9469	−1.1	−0.4	−0.0011
Low	2.5	3	68.7	59.9	0.9922	67.5	61.6	0.9919	−1.2	+1.7	−0.0003
Low	2.5	4	93.5	92.7	0.9989	93.2	92.6	0.9989	−0.2	−0.1	−0.0001
Low	2.5	6	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	2.5	8	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	2.5	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	2.5	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	5	1	10.9	8.6	0.8780	10.3	6.9	0.8746	−0.7	−1.8	−0.0034
Low	5	2	57.4	66.5	0.9877	56.9	63.5	0.9870	−0.5	−3.0	−0.0006
Low	5	3	96.7	96.4	0.9997	96.2	95.8	0.9997	−0.5	−0.5	−0.0000
Low	5	4	100.0	100.0	1.0000	99.9	100.0	1.0000	−0.1	+0.1	+0.0000
Low	5	6	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	5	8	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	5	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	5	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	10	1	20.4	20.2	0.9440	19.6	18.9	0.9405	−0.8	−1.4	−0.0034
Low	10	2	90.8	90.6	0.9987	89.8	90.2	0.9985	−1.1	−0.4	−0.0002
Low	10	3	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	−0.0000
Low	10	4	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	10	6	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	10	8	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Low	10	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000

Table S3 continued

Noise	Depth (M)	FF (%)	Sens.	Sens. FPR	AUC	Sens.	Sens. FPR	AUC	Δ Sens.	Δ Sens. FPR	Δ AUC
Low	10	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	1	1	1.1	1.2	0.6901	1.0	1.0	0.6877	−0.1	−0.2	−0.0023
Realistic	1	2	6.7	5.2	0.8426	6.8	4.6	0.8407	+0.1	−0.6	−0.0019
Realistic	1	3	18.9	17.6	0.9325	17.9	19.1	0.9304	−0.9	+1.4	−0.0021
Realistic	1	4	41.2	40.7	0.9753	40.0	39.9	0.9746	−1.2	−0.8	−0.0007
Realistic	1	6	87.0	81.9	0.9984	86.7	82.2	0.9984	−0.3	+0.2	−0.0000
Realistic	1	8	99.1	98.7	0.9999	99.1	98.6	0.9999	+0.0	−0.1	−0.0000
Realistic	1	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	1	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	2.5	1	3.1	1.7	0.7794	2.9	1.2	0.7745	−0.2	−0.5	−0.0049
Realistic	2.5	2	19.7	17.9	0.9351	18.9	17.5	0.9317	−0.7	−0.4	−0.0033
Realistic	2.5	3	61.3	59.9	0.9871	59.1	59.0	0.9860	−2.2	−0.9	−0.0011
Realistic	2.5	4	87.6	85.9	0.9981	86.4	84.9	0.9978	−1.2	−1.0	−0.0002
Realistic	2.5	6	100.0	100.0	1.0000	99.9	100.0	1.0000	−0.1	+0.0	−0.0000
Realistic	2.5	8	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	2.5	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	2.5	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	5	1	6.1	7.1	0.8322	5.6	6.9	0.8245	−0.5	−0.3	−0.0077
Realistic	5	2	43.9	43.0	0.9785	41.0	40.6	0.9758	−2.9	−2.4	−0.0028
Realistic	5	3	86.1	87.7	0.9979	84.4	86.6	0.9974	−1.7	−1.1	−0.0004
Realistic	5	4	99.0	99.6	0.9999	98.7	99.2	0.9999	−0.3	−0.4	−0.0000
Realistic	5	6	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	5	8	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	5	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	5	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	10	1	13.1	10.4	0.8962	11.2	10.0	0.8853	−1.9	−0.4	−0.0109
Realistic	10	2	67.0	64.8	0.9919	62.6	60.0	0.9896	−4.4	−4.8	−0.0023
Realistic	10	3	98.5	98.0	0.9998	97.2	96.9	0.9997	−1.2	−1.1	−0.0001
Realistic	10	4	100.0	100.0	1.0000	100.0	100.0	1.0000	−0.1	+0.0	−0.0000
Realistic	10	6	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	10	8	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	10	10	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000
Realistic	10	15	100.0	100.0	1.0000	100.0	100.0	1.0000	+0.0	+0.0	+0.0000

Table S4. Detection limits of W with true-FF and fixed-FF weights. Main and conservative (Wilson lower bound) detection limits of W with weights from the true FF and with weights fixed at FF 10%, for each depth and noise level. The main limits differ by at most 0.08 percentage points.

Noise	Depth (M)	Main, true FF	Main, fixed FF	Conservative, true FF	Conservative, fixed FF
None	1	7.03%	7.05%	7.21%	7.22%
None	2.5	4.52%	4.50%	4.76%	4.74%
None	5	2.94%	2.95%	2.97%	2.98%
None	10	1.99%	2.00%	2.07%	2.15%
Low	1	7.18%	7.18%	7.32%	7.33%
Low	2.5	4.46%	4.52%	4.71%	4.76%
Low	5	2.96%	2.97%	2.98%	2.99%
Low	10	2.45%	2.51%	2.53%	2.58%
Realistic	1	7.33%	7.34%	7.46%	7.47%
Realistic	2.5	5.20%	5.28%	5.31%	5.38%
Realistic	5	3.69%	3.74%	3.76%	3.80%
Realistic	10	2.89%	2.93%	2.91%	2.96%

Table S5. AUC of all methods in all conditions. Area under the ROC curve of methods A, B, B-opt, W, C1 and C2 on the test group (10,000 euploid and 2,000 trisomic patients per condition), rounded to four decimals.

Noise	Depth (M)	FF (%)	A	B	B-opt	W	C1	C2
None	1	1	0.6509	0.6644	0.6641	0.6972	0.6423	0.6817
None	1	2	0.7841	0.8075	0.8054	0.8534	0.7773	0.8483
None	1	3	0.8800	0.8994	0.8964	0.9379	0.8742	0.9306
None	1	4	0.9388	0.9541	0.9524	0.9799	0.9333	0.9765
None	1	6	0.9905	0.9939	0.9936	0.9984	0.9860	0.9964
None	1	8	0.9992	0.9995	0.9996	0.9999	0.9980	0.9993
None	1	10	0.9999	1.0000	1.0000	1.0000	0.9991	0.9994
None	1	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
None	2.5	1	0.7337	0.7547	0.7552	0.8043	0.7281	0.7992
None	2.5	2	0.8879	0.9152	0.9163	0.9539	0.8845	0.9496
None	2.5	3	0.9669	0.9784	0.9781	0.9926	0.9647	0.9879
None	2.5	4	0.9935	0.9962	0.9956	0.9992	0.9926	0.9938
None	2.5	6	0.9999	1.0000	1.0000	1.0000	0.9985	0.9997
None	2.5	8	1.0000	1.0000	1.0000	1.0000	0.9991	1.0000
None	2.5	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
None	2.5	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
None	5	1	0.7950	0.8298	0.8302	0.8717	0.7889	0.8642
None	5	2	0.9604	0.9750	0.9747	0.9908	0.9570	0.9869
None	5	3	0.9956	0.9969	0.9968	0.9994	0.9917	0.9963
None	5	4	0.9998	0.9999	0.9999	1.0000	0.9992	0.9990
None	5	6	1.0000	1.0000	1.0000	1.0000	0.9998	1.0000
None	5	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
None	5	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
None	5	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
None	10	1	0.8844	0.9189	0.9174	0.9533	0.8802	0.9500
None	10	2	0.9934	0.9971	0.9970	0.9996	0.9911	0.9992
None	10	3	0.9999	1.0000	1.0000	1.0000	0.9996	0.9992
None	10	4	1.0000	1.0000	1.0000	1.0000	0.9996	1.0000
None	10	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
None	10	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
None	10	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
None	10	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Low	1	1	0.6437	0.6707	0.6711	0.6978	0.6401	0.6911
Low	1	2	0.7764	0.8138	0.8142	0.8514	0.7731	0.8442
Low	1	3	0.8753	0.8948	0.8930	0.9331	0.8711	0.9278
Low	1	4	0.9383	0.9534	0.9541	0.9794	0.9371	0.9763
Low	1	6	0.9903	0.9944	0.9944	0.9990	0.9892	0.9977
Low	1	8	0.9990	0.9996	0.9996	1.0000	0.9979	0.9995
Low	1	10	1.0000	1.0000	1.0000	1.0000	0.9995	1.0000
Low	1	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Low	2.5	1	0.7132	0.7484	0.7434	0.7891	0.7082	0.7819
Low	2.5	2	0.8844	0.9171	0.9161	0.9481	0.8826	0.9415
Low	2.5	3	0.9633	0.9775	0.9780	0.9922	0.9614	0.9877
Low	2.5	4	0.9897	0.9957	0.9955	0.9989	0.9854	0.9953
Low	2.5	6	0.9998	1.0000	0.9999	1.0000	0.9959	0.9995
Low	2.5	8	1.0000	1.0000	1.0000	1.0000	0.9999	1.0000
Low	2.5	10	1.0000	1.0000	1.0000	1.0000	0.9998	1.0000
Low	2.5	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Low	5	1	0.7970	0.8374	0.8383	0.8780	0.7894	0.8732
Low	5	2	0.9485	0.9711	0.9710	0.9877	0.9461	0.9837
Low	5	3	0.9926	0.9982	0.9983	0.9997	0.9878	0.9978
Low	5	4	0.9996	0.9999	0.9999	1.0000	0.9988	0.9994
Low	5	6	1.0000	1.0000	1.0000	1.0000	0.9992	1.0000
Low	5	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Low	5	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Low	5	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table S5 continued

Noise	Depth (M)	FF (%)	A	B	B-opt	W	C1	C2
Low	10	1	0.8642	0.9147	0.9152	0.9440	0.8584	0.9409
Low	10	2	0.9833	0.9958	0.9957	0.9987	0.9790	0.9955
Low	10	3	0.9993	1.0000	1.0000	1.0000	0.9978	0.9999
Low	10	4	1.0000	1.0000	1.0000	1.0000	0.9995	1.0000
Low	10	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Low	10	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Low	10	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Low	10	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Realistic	1	1	0.6419	0.6547	0.6522	0.6901	0.6360	0.6802
Realistic	1	2	0.7701	0.7984	0.7986	0.8426	0.7639	0.8365
Realistic	1	3	0.8568	0.8992	0.9007	0.9325	0.8525	0.9285
Realistic	1	4	0.9266	0.9549	0.9537	0.9753	0.9241	0.9708
Realistic	1	6	0.9869	0.9935	0.9932	0.9984	0.9803	0.9967
Realistic	1	8	0.9980	0.9994	0.9994	0.9999	0.9966	0.9987
Realistic	1	10	0.9999	1.0000	0.9999	1.0000	0.9998	0.9987
Realistic	1	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Realistic	2.5	1	0.7025	0.7483	0.7459	0.7794	0.6979	0.7756
Realistic	2.5	2	0.8566	0.9093	0.9097	0.9351	0.8533	0.9317
Realistic	2.5	3	0.9438	0.9744	0.9743	0.9871	0.9376	0.9842
Realistic	2.5	4	0.9806	0.9949	0.9950	0.9981	0.9749	0.9937
Realistic	2.5	6	0.9991	1.0000	0.9999	1.0000	0.9962	0.9987
Realistic	2.5	8	1.0000	1.0000	1.0000	1.0000	0.9989	1.0000
Realistic	2.5	10	1.0000	1.0000	1.0000	1.0000	0.9999	1.0000
Realistic	2.5	15	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998
Realistic	5	1	0.7355	0.8136	0.8116	0.8322	0.7324	0.8325
Realistic	5	2	0.9129	0.9676	0.9695	0.9785	0.9109	0.9744
Realistic	5	3	0.9751	0.9961	0.9955	0.9979	0.9703	0.9938
Realistic	5	4	0.9962	0.9998	0.9998	0.9999	0.9947	0.9988
Realistic	5	6	1.0000	1.0000	1.0000	1.0000	0.9997	1.0000
Realistic	5	8	1.0000	1.0000	1.0000	1.0000	0.9997	1.0000
Realistic	5	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Realistic	5	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Realistic	10	1	0.7819	0.8872	0.8952	0.8962	0.7741	0.9044
Realistic	10	2	0.9402	0.9908	0.9923	0.9919	0.9378	0.9900
Realistic	10	3	0.9890	0.9999	0.9999	0.9998	0.9847	0.9973
Realistic	10	4	0.9988	1.0000	1.0000	1.0000	0.9972	0.9997
Realistic	10	6	1.0000	1.0000	1.0000	1.0000	0.9992	1.0000
Realistic	10	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Realistic	10	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Realistic	10	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table S6. Detection limits with conservative and matched-FPR estimates. For each method, depth and noise level: the main detection limit (sensitivity at each method's own threshold, as in Table 2 of the main text), the conservative limit (Wilson 95% lower bound of the sensitivity) and the limit from the sensitivity at FPR 0.135%. No limit at matched FPR differed from the main limit by more than one grid step.

Noise	Depth (M)	Method	Main	Conservative	Matched FPR
None	1	A	8.94%	9.13%	8.87%
None	1	B	7.92%	8.00%	8.07%
None	1	B-opt	7.93%	8.05%	8.00%
None	1	W	7.03%	7.21%	7.31%
None	1	C1	9.03%	9.27%	8.89%
None	1	C2	7.91%	7.92%	7.82%
None	2.5	A	5.78%	5.83%	5.79%
None	2.5	B	5.61%	5.66%	5.58%
None	2.5	B-opt	5.61%	5.66%	5.57%
None	2.5	W	4.52%	4.76%	4.64%
None	2.5	C1	5.84%	5.89%	5.83%
None	2.5	C2	5.90%	5.90%	5.85%
None	5	A	3.90%	3.94%	3.92%
None	5	B	3.74%	3.78%	3.69%
None	5	B-opt	3.73%	3.78%	3.71%
None	5	W	2.94%	2.97%	2.94%
None	5	C1	3.98%	3.99%	3.97%
None	5	C2	3.95%	3.95%	3.79%
None	10	A	2.88%	2.90%	2.88%
None	10	B	2.73%	2.77%	2.75%
None	10	B-opt	2.74%	2.78%	2.69%
None	10	W	1.99%	2.07%	1.99%
None	10	C1	2.91%	2.94%	2.90%
None	10	C2	2.00%	2.17%	1.99%
Low	1	A	9.06%	9.24%	8.48%
Low	1	B	7.94%	8.07%	8.19%
Low	1	B-opt	8.04%	8.43%	8.36%
Low	1	W	7.18%	7.32%	7.06%
Low	1	C1	9.38%	9.50%	8.62%
Low	1	C2	7.35%	7.49%	7.15%
Low	2.5	A	5.78%	5.83%	5.83%
Low	2.5	B	5.49%	5.56%	5.59%
Low	2.5	B-opt	5.51%	5.58%	5.57%
Low	2.5	W	4.46%	4.71%	4.63%
Low	2.5	C1	5.87%	5.91%	5.85%
Low	2.5	C2	5.90%	5.91%	4.74%
Low	5	A	4.61%	4.83%	3.99%
Low	5	B	3.70%	3.76%	3.69%
Low	5	B-opt	3.69%	3.75%	3.69%
Low	5	W	2.96%	2.98%	2.95%
Low	5	C1	4.13%	4.48%	4.00%
Low	5	C2	3.38%	3.48%	3.10%
Low	10	A	2.99%	3.13%	3.08%
Low	10	B	2.77%	2.80%	2.81%
Low	10	B-opt	2.77%	2.80%	2.82%
Low	10	W	2.45%	2.53%	2.47%
Low	10	C1	3.95%	3.95%	3.85%
Low	10	C2	2.95%	2.95%	2.85%
Realistic	1	A	9.42%	9.60%	9.36%
Realistic	1	B	8.32%	8.67%	8.33%
Realistic	1	B-opt	8.55%	8.85%	8.50%
Realistic	1	W	7.33%	7.46%	7.56%
Realistic	1	C1	9.46%	9.63%	9.41%
Realistic	1	C2	7.94%	7.95%	7.88%

Table S6 continued

Noise	Depth (M)	Method	Main	Conservative	Matched FPR
Realistic	2.5	A	6.86%	7.04%	6.89%
Realistic	2.5	B	5.71%	5.77%	5.70%
Realistic	2.5	B-opt	5.71%	5.77%	5.67%
Realistic	2.5	W	5.20%	5.31%	5.30%
Realistic	2.5	C1	7.39%	7.48%	6.95%
Realistic	2.5	C2	5.90%	5.91%	5.56%
Realistic	5	A	5.61%	5.67%	5.60%
Realistic	5	B	3.86%	3.90%	3.84%
Realistic	5	B-opt	3.85%	3.90%	3.85%
Realistic	5	W	3.69%	3.76%	3.62%
Realistic	5	C1	5.91%	5.92%	5.83%
Realistic	5	C2	3.72%	3.79%	3.67%
Realistic	10	A	4.60%	4.82%	4.67%
Realistic	10	B	2.91%	2.94%	2.91%
Realistic	10	B-opt	2.90%	2.92%	2.92%
Realistic	10	W	2.89%	2.91%	2.91%
Realistic	10	C1	5.17%	5.29%	5.06%
Realistic	10	C2	2.80%	2.84%	2.92%